

Layered, Evidence-Aware Bluetooth Fingerprinting: Low-Cost Signals and Conflict-Aware Inference

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Abstract—We infer the type and operating system of a nearby Bluetooth device without changing that device’s state. Knowing the operating system supports security auditing, because many documented weaknesses are specific to particular versions, so an OS label narrows which of them could apply. Existing methods each reach part of this: they flood the target, read a single protocol field, or only listen to the radio, and each falls short in some respect. We aim instead for three properties together — low interaction cost, the ability to distinguish many operating systems, and reporting conflicts between signals rather than hiding them. We extend an existing baseline scanner with additional signal layers obtained at low interaction cost, add one cache-read signal usable on re-encounter, and attach an *evidence chain* that leaves the prediction unchanged but labels each signal relative to it and lowers the reported confidence on a genuine conflict. We then measure how far that mechanism works on real data, which produced our main finding. Both signal pairs the chain originally compared depended on GATT Appearance, present in 2 of 149 records and 0 of 77 scans, so the chain never fired; a pair requiring no Appearance raised coverage from 0% to 48% of scans, but to 62% on our own devices against 10% on third-party ones. What limits this class of inference is therefore not the quality of a signal but the simultaneous availability of the signals a check requires. We claim no calibrated accuracy result, and present this as a measurement that openly reports a constraint we observed.

Index Terms—Bluetooth fingerprinting, BLE privacy, operating-system identification, evidence-based inference, uncertainty, security audit.

I. INTRODUCTION

Bluetooth is present in phones, wearables, peripherals, and IoT devices almost everywhere. The focus of this work is to infer the type and operating system of a nearby Bluetooth device without changing the device’s state. We build on a baseline active scanner developed earlier within our group.

The value of doing so is auditing. If we know a device’s operating system, we can audit its security, and since various vulnerabilities are specific to certain versions, an OS indicator considerably narrows down which vulnerabilities of that device could be exploited [5]. Bluetooth was never designed to expose such an identifier, and privacy mechanisms such as MAC-address randomization work against recovering one [4].

Existing detection methods almost do this job, but each of them falls short in some respect. Flooding-based probing recovers an OS version by observing the target under stress [1], at a high interaction cost and with no representation of uncertainty. Methods that read a single protocol field or a set

of service metadata are cheap [2] but stop short of an operating system. Methods that only listen to the radio are passive and hard to spoof [3] but need special hardware and return an identity rather than an OS.

This work aims for three desirable properties together: low interaction cost, the ability to distinguish many operating systems, and reporting conflicts between signals rather than hiding them. In other words, rather than producing a new signal, the aim is both to enrich what is already available and to make uncertainties visible.

Our four contributions are:

- **Added signal layers at low interaction cost** (Section III-A): the LMP feature bitmap, chipset and manufacturer resolution, GATT Appearance, GATT connection parameters, and the full SDP attribute tree.
- **A cache-read signal usable on re-encounter** (Section III-A): Modalias, read from the host cache rather than from the device.
- **An evidence chain for conflict-aware inference** (Section III-B): each signal is labelled relative to the inference, and genuine disagreement is reported rather than averaged away.
- **A measurement of how, and to what extent, the mechanism works on real data** (Section V).

Constraints and non-goals. The most fundamental constraint is that every signal this method uses — Class-of-Device, device name, GATT Appearance, Modalias, SDP — is declared by the target device and is not cryptographically bound. The device may declare them falsely and mislead us, and we know of no existing method to detect this. In addition, MAC randomization causes the same device to be seen as multiple addresses; our sample is small, so the rates we measure carry wide confidence intervals; and the confidence score we produce is not calibrated. Accordingly, this work infers a device type and an operating-system family, but does not claim to extract an exact OS version number, does not break MAC randomization, and does not perform an accuracy comparison with other methods on shared data. Such a comparison would require a ground-truth dataset, a layer-by-layer ablation, and a calibrated confidence function. What we actually measure here, separately from all of these, is the availability of the signals.

II. BACKGROUND

It is worth being precise about what the existing families of work do, since the contribution here is not a better signal but a fuller use of the ones already within reach.

Flooding the target: Kavisankar *et al.* [1] The closest work in intent identifies the operating system of a Bluetooth device by sending traffic and observing how the device responds. This reaches a fine-grained answer, and it pays for it twice: the target’s state is disturbed, which is exactly what we set out to avoid, and the result arrives as a single answer with no account of how well supported it was.

Reading a single protocol field: Celosia and Cunche [2] Fingerprinting from the Generic Attribute Profile a BLE device exposes shows that declared metadata alone is often enough to name a device. This is cheap and does not disturb the device, and we borrow from it directly. Its limit is scope: a device model is not an operating-system family, and a method resting on one field inherits whatever that field happens to contain.

Only listening to the radio: Santana-Cruz *et al.* [3] Identifying a device from imperfections in its radio emissions is the hardest of the three to spoof, because the device is never asked to declare anything. It is also the least deployable, requiring dedicated hardware, and it recognises *that* device again rather than reporting what it runs.

Our position. We sit with the declarative methods and accept their central weakness, stated above: every signal is declared by the target. What we add is breadth, several declared signals rather than one, and an explicit account of what happens when those signals do not agree with each other.

III. APPROACH

A. Layered structure

The system scans a single target address in layers. Each layer is defined by what it costs to obtain rather than by what it reveals.

Layer 0: the baseline active scan. The inherited scanner, which reads the Class-of-Device, the SDP service list, the GATT service identifiers, the vendor prefix of the address (OUI), and the LMP protocol version.

Layer 1: parsing what is already there, and one cache read. The baseline already issues the query whose response carries the LMP feature bitmap; we parse that bitmap rather than discarding it, so the signal costs parsing and no new connection. Layer 1 also reads *Modalias* from the BlueZ cache, and resolves chipset and manufacturer from it where present. Reading the cache requires no live connection at scan time. It does require that the host has previously interacted with the device, so *Modalias* is a signal for a re-encounter or a re-scan rather than a generally free one. We state this as a scope limit here, and Section V confirms it empirically rather than assuming it.

Layer 2: BLE category and the full SDP tree. We read GATT Appearance, which matters especially for BLE-only devices that expose no Class-of-Device, together with the GATT connection parameters; and where the baseline reads

TABLE I
ACCESS PATH PER SIGNAL. THE BR/EDR SIGNALS REUSE THE CONNECTION THE BASELINE ALREADY OPENS; GATT IS READ OVER A SEPARATE LE TRANSPORT; THE CACHE READ NEEDS NO LIVE CONNECTION AT SCAN TIME.

Signal	Access path
LMP version and feature bitmap	BR/EDR (reuses baseline)
SDP service list and attribute tree	BR/EDR (reuses baseline)
Class-of-Device	inquiry / host cache
GATT Appearance, conn. params, UUIDs	LE (separate transport)
Modalias	host cache (no live conn.)
RSSI / RTT (Layer 3)	active observation window

only a summarised service list, we read the full SDP attribute tree.

One point here matters more than its technical detail suggests: *these signals do not share a single connection.* SDP goes over the BR/EDR connection the baseline already opens, but GATT is read over a separate LE transport. A device that answers on one may be silent on the other, and Table I gives the access path per signal for that reason. It is the table that explains most of the availability results later.

Layer 3: time-windowed behavioural observation. A short windowed observation such as RSSI and RTT. It is measured and reported, but it does not feed back into the current inference. We state this explicitly because it bounds every claim that follows: no inference in this work relies on this layer.

B. Evidence chain

The methodological core of this work is not a signal but a discipline about how signals are combined.

The evidence chain is an audit layer produced separately, after the inference has already been determined. It does not change the inference. What it does is label each signal as *primary*, *supports*, *conflicts* or *unused* relative to that inference, and lower the reported confidence in the case of a genuine conflict. This is the point of departure from a single fused score: if two signals point in opposite directions, a score records only their sum, and the operator never learns that the device contradicted itself.

Avoiding circularity. The *primary* label carries the idea that makes the rest work. If an inference was produced by a signal, that signal is labelled *primary* and excluded from the count, because comparing it with the result it derived itself would be self-validation. Only signals that played no part in producing the inference are eligible to support or contradict it. The same argument recurs throughout this design, and it is the reason several things one might expect to find here are deliberately absent.

Two stages. The process takes place in two stages. First, each signal is evaluated against the inference. Second, pairs of independently produced signals are evaluated against each other rather than against the inference. A pair is evaluated only when both of its halves resolved: a device on which only one half is present yields no record, since an absent check is not

a passed check. That property is unremarkable in itself, and it turns out to govern the central result of Section V.

C. Re-encounter comparison

When a device is scanned twice, the question that comes to mind is whether this scan is consistent with the previous one. The obvious way of answering it — comparing the two inferences — is closed to us by the circularity argument above. Agreement between two inferences measures not accuracy but the system’s internal consistency, because both inferences come from the same scorer.

Therefore, in this part we compare only the invariant fields: Modalias, Class-of-Device, and the OUI-resolved manufacturer. These fields cannot differ within a short interval; if they differ, there is an error in at least one place, and that conclusion is available without any ground truth. A disagreement is recorded as *conflicts*. An agreement is counted as *unused*, because a reader agreeing with itself is not independent corroboration. A device scanned for the first time does not enter this comparison at all.

D. Scoring model

The device type is determined by a branching system with a fixed priority order, in which the first match wins: Apple, Android, embedded IoT, PC/laptop, wearable, audio. Only the Android branch has a numerical model; the remaining branches are rule-based. We describe the model at this level deliberately, since the weights it uses are not calibrated (Section III) and presenting them as tuned parameters would overstate what they are.

E. Worked examples

The mechanisms above are easier to judge against concrete records than in the abstract. Table II lists every distinct device in the full-scan archive — 26 devices over 87 scans — replayed against the current scorer, and groups them by what the evidence chain was able to do with each.

The grouping is itself the result. The pair of Section III-B could be evaluated on 9 of the 26 devices; on the remaining 17 at least one half did not resolve, so no record was produced and the chain was empty. Rows 1 and 11 show why that matters: they are the same physical earbuds, with the same cached Modalias and the same Class-of-Device, distinguished only by whether the advertised name resolved. With the name, the cascade returns an audio peripheral running an embedded firmware; without it, the same device is reported as an Apple phone or computer running iOS 17+. Both answers carry confidence *High*, and in the second the chain has nothing to say, because an absent check is not a passed check.

Rows 8 and 9 state a limitation of the two-stage design that we had not identified before assembling this table. On the smartwatch and on the Samsung earbuds, the two category signals agree with each other — wearable and wearable, audio and audio — so each pair is recorded as *supports*. The prediction, however, is *Smartphone* in both cases, contradicting both halves of the pair it was just told agree. Nothing in

the chain catches this: the second stage compares the halves *to each other* and never to the prediction, while the first stage does not cover these signals at all, since they fed the cascade and are therefore *primary* and excluded by the circularity rule of Section III-B. The rule exists so that a signal cannot corroborate a prediction it produced; in excluding those signals it also gives up the ability to notice that the prediction disagrees with all of them at once. A third check — comparing an agreeing pair against the prediction and flagging the case where it matches neither half — would catch both rows, and we state it as future work rather than as something this system does.

Two further patterns in the table bear on the availability argument of Section V, and neither is flattering. First, rows 12 and 16: the OUI resolves the vendor as Apple, the cached Modalias resolves the vendor as Apple, and the prediction is Android 15 at confidence *High*. Here the contradicting evidence *was* available and no stage of the chain looks at it, because no check compares a resolved vendor against a predicted OS family. That is a missing check rather than a missing signal, and it is a different failure from the co-availability one this paper argues for.

Second, the largest group in the table is the 13 devices that advertised a name which our derivation mapped to no category. Several of those names — *realme 11 Pro+ 5G*, *OPPO A5 2020*, and two Samsung flagships named after their model — are perfectly serviceable product strings. The pair did not fire on them because our keyword table did not cover them, not because the device withheld anything. Co-availability, as we measure it, is therefore partly a property of the environment and partly an artefact of our own derivation, and the 48% coverage figure of Section V should be read with that split in mind.

F. Differences from the baseline

In the baseline work, every signal contributed to a single weighted score and a single decision was made without uncertainty, and Modalias was not read. We added two things: the new signal layers of Section III-A and the evidence chain of Section III-B. The result is not a new detector. With almost zero new connections, it uses the resources already at hand and turns the system into one in which disagreement between signals is reported.

IV. COMPARISON WITH EXISTING METHODS

The methods of Section II are evaluated on different device sets with different mechanisms, so their reported accuracies are not comparable and we do not tabulate them as if they were; as stated in the introduction, an accuracy comparison on shared data is explicitly not something this work performs. What can be compared is what each method *requires*, what it *emits*, and how it behaves when its evidence is weak. Table III sets that out along the three properties this work set out to obtain together.

Read across the rows, the table states the claim of the introduction in a compact form. Each existing family obtains

TABLE II

EVERY DISTINCT DEVICE IN THE FULL-SCAN ARCHIVE (26 DEVICES, 87 SCANS), REPLAYED AGAINST THE CURRENT SCORER AND GROUPED BY WHAT THE EVIDENCE CHAIN WAS ABLE TO DO. **CoD** IS THE CLASS-OF-DEVICE MAJOR CLASS; **NAME CAT.** IS THE MAJOR CLASS IMPLIED BY THE ADVERTISED NAME; THE PAIR OF SECTION III-B IS EVALUATED ONLY WHEN BOTH RESOLVE. **Mod.** IS THE MODALIAS VENDOR AND PRODUCT PREFIX, **LMP** THE LINK-LAYER VERSION. NAMES CHOSEN BY THEIR OWNERS ARE REDACTED AS **<NAME>**; THE REST ARE PRODUCT STRINGS THE DEVICES ADVERTISE. PREDICTIONS IN BOLD ARE THOSE WHERE THE VENDOR IS RESOLVED AS APPLE AND THE PREDICTED OS IS ANDROID. EVERY PREDICTION HERE IS REPORTED AT CONFIDENCE *High* EXCEPT THE TWO *Unknown Peripheral* ROWS, WHICH ARE *Low*.

	Advertised name	Scans	CoD	name cat.	Mod.	LMP	Vendor	Prediction	Chain
<i>Pair evaluated; prediction consistent with it</i>									
1	<name> (AirPods)	13	0x04	audio	v004Cp201	14	Apple	Audio Peripheral (Apple), Embedded Firmware (RTOS) <i>Audio Sink Device (LMP 14)</i>	supports
2	<name> iPhone	11	0x02	phone	v004Cp741	12	Apple	Smartphone/Computer (Apple), iOS / macOS <i>iOS 16.x / macOS 13.x</i>	supports
3	Redmi A5	2	0x02	phone	v00E0p000	11	Xiaomi	Smartphone, Android <i>Android 15 (Vanilla Ice Cream)</i>	supports
4	<name>.iPhone	1	0x02	phone	v004Cp741	11	Apple	Smartphone (iPhone), iOS <i>iOS 15+</i>	supports
5	iPad von <name>	1	0x01	computer	v004Cp770	12	Apple	Smartphone/Computer (Apple), iOS / macOS <i>iOS 16.x / macOS 13.x</i>	supports
6	iPad (2)	1	0x01	computer	v004Cp790	12	Apple	Smartphone/Computer (Apple), iOS / macOS <i>iOS 16.x / macOS 13.x</i>	supports
7	<name> iPad	1	0x01	computer	v004Cp761	12	Apple	Computer (Mac), macOS <i>macOS 13+</i>	supports
<i>Pair evaluated; prediction contradicts both halves of the pair</i>									
8	HUAWEI WATCH FIT 3-8CF	8	0x07	wearable	–	11	Huawei	Smartphone, Android <i>Android 13 (Tiramisu)</i>	supports
9	Galaxy Buds2 (C7D2)	5	0x04	audio	v0075pA01	11	Samsung	Smartphone, Android <i>Android 15 (Vanilla Ice Cream)</i>	supports
<i>No pair: Class-of-Device present, advertised name resolved to no category</i>									
10	realme 11 Pro+ 5G	12	0x02	–	v00E0p000	12	Realme	Smartphone, Android <i>Android 15 (Vanilla Ice Cream)</i>	empty
11	(none)	5	0x04	–	v004Cp201	14	Apple	Smartphone/Computer (Apple), iOS / macOS <i>iOS 17+ / macOS 14+</i>	empty
12	<name>	4	0x02	–	v004Cp750	12	Apple	Smartphone, Android <i>Android 15 (Vanilla Ice Cream)</i>	empty
13	OPPO A5 2020	2	0x02	–	v001Dp120	9	OPPO	Smartphone, Android <i>Android 14 (Upside Down Cake)</i>	empty
14	(none)	2	0x04	–	v0075pA01	–	Samsung	Unknown Peripheral, Unknown	empty
15	(none)	1	0x02	–	v00E0p000	–	Realme	Unknown Peripheral, Unknown	empty
16	<name>	1	0x02	–	v004Cp760	12	Apple	Smartphone, Android <i>Android 15 (Vanilla Ice Cream)</i>	empty
17	S22 Ultra von <name>	1	0x02	–	v0075p010	11	Samsung	Smartphone, Android <i>Android 15 (Vanilla Ice Cream)</i>	empty
18	S25+ von <name>	1	0x02	–	v0075p010	13	Samsung	Smartphone, Android <i>Android 15 (Vanilla Ice Cream)</i>	empty
19	(none)	1	0x01	–	v0006p000	8	Intel	PC / Laptop, Windows <i>Windows 8.1 / 10</i>	empty
20	<name>	1	0x01	–	v0006p000	13	Liteon	PC / Laptop, Windows <i>Windows 11</i>	empty
21	“a (2)”	1	0x02	–	–	14	Apple	Smartphone (iPhone), iOS <i>iOS 17+</i>	empty
22	<name>	1	0x02	–	v004Cp761	12	Apple	Smartphone (iPhone), iOS <i>iOS 16+</i>	empty
<i>No pair: name category present, no Class-of-Device</i>									
23	<name>-LAPTOP	1	–	computer	–	9	Fugui	PC / Laptop, Windows <i>Windows 8.1 / 10</i>	empty
<i>No pair: neither half resolved</i>									
24	(none)	8	–	–	–	13	Liteon	PC / Laptop, Windows <i>Windows 11</i>	empty
25	<name>	1	–	–	–	13	Liteon	PC / Laptop, Windows <i>Windows 11</i>	empty
26	<name>’s S22	1	–	–	–	12	Samsung	Smartphone, Android <i>Android 15 (Vanilla Ice Cream)</i>	empty

two of the three properties at most. The flooding method reaches an OS version but disturbs the target and reports one verdict whatever the evidence behind it; the declarative metadata methods are cheap and quiet but stop at a device model; the radio methods ask the device to declare nothing at all, which makes them the hardest to mislead, but they need dedicated hardware and return an identity rather than an operating system. The baseline we extend reaches an OS but folds every signal into one number, so a disagreement between two of its inputs leaves no trace in its output.

The last row is not a claim of superiority, and one column is deliberately unflattering. This work obtains the three properties together, but the conflict reporting it adds is only as available as the signals a check needs — which is what Section V measures, and where it arrives at 48% of scans overall and 10% on third-party devices. A mark in the final column therefore describes a mechanism that exists, not one that fires on every scan.

V. PRELIMINARY SIGNAL-AVAILABILITY MEASUREMENT

The question this section answers is how often the Layer 1–2 signals we added are actually present in the surroundings. This is not an accuracy study, and we treat it as the prior question rather than a lesser one: a rarely present signal, no matter how well it classifies, cannot help us as long as it is not present. In this sense availability took precedence over accuracy.

Method. A standalone audit passively discovers Bluetooth addresses and then runs only the Layer 1–2 queries against each, recording per device whether it was *reached* at all and, if so, which signals it exposed. We separate availability from the signal being rich: a device that opens a connection but exposes no tracked signal is counted as reached, not as a connection failure, since conflating the two would make the fraction of reached devices exposing a given signal partly circular. Two availability rounds were run: a pilot in a single office and a wider sweep across several locations. Separately, a set of full scans was archived, and those archives are what the evidence-chain measurement below replays.

Reachability is the first constraint. The pipeline could not reach the vast majority of addresses at all: 16 of 18 in the pilot, and 46 of 52 in the sweep (Table IV). Whatever the added signals are worth, they are worth nothing on an address the pipeline never opens a channel to, and this dominates every other availability figure we report.

Modalias is a re-scan signal, and the measurement shows it. Modalias came out at 44% on previously paired devices in the office, while it was not found at all in scans at different locations. The pilot ran where the host had already dealt with those devices; the sweep met unfamiliar ones and found nothing cached. This is the scope claim of Section III-A turned into evidence rather than left as an assurance: Modalias is only useful on re-scan.

The evidence chain did not work on real data, and the reason is the finding. Having built the chain, we examined how it works on the archived scans, and found that it did not

fire. The cause was not in its logic. Both of the first two pairs we had chosen depended on GATT Appearance, and this signal was present in only 2 of 149 records and in 0 of 77 scans. We had gated the entire mechanism on the scarcest signal in the study.

The error is instructive because of how it was made. We had chosen the pairs according to how *independent* they were, and had not paid attention to how often they are present *together*. Choosing a new pair that does not require Appearance — Class-of-Device against the advertised name — raised coverage from 0% to 48% of scans.

Coverage does not transfer to unfamiliar devices. Table V carries the result we consider most important, and it is the least flattering one here. The method worked at 62% on our own devices and at only 10% on third-party devices. The own-device figure is therefore an upper bound that does not transfer to the setting the method exists for, and reporting it alone would have overstated the method’s reach by roughly a factor of six.

What this means for the design. What limits this kind of inference is not the quality of the signal but the simultaneous availability — the co-availability — of the required signals. A pair is useful only if both halves are also commonly present, and we can draw a further conclusion from this: independence and availability work against each other. The signals that are most clearly independent of the primary decision path are precisely the ones that path did not need, and therefore the ones a device is least likely to expose. At our sample sizes we state this as the constraint that governs how such a mechanism should be built, and as a hypothesis the pair change above is designed to test, rather than as a settled result.

Limitations of this measurement. It measures availability, not identification accuracy. The sample is small, so every proportion carries a wide confidence interval, and the rounds count *addresses* rather than distinct devices, which MAC randomization means is not the same thing. The evidence chain, though it now fires, has not yet produced a genuine device-originated conflict on real data, and Section III-E shows a class of disagreement — the prediction contradicting an agreeing pair — that its current two stages cannot express at all.

VI. CONCLUSION

In this work we added low-cost new signal layers, and an evidence chain that makes conflicts visible, to an existing baseline scanner. The goal was not so much to produce a new signal as to enrich the resources at hand and report uncertainty honestly.

When we measured the mechanism on real data, we learned the actual lesson. What limits this kind of inference is not the quality of the signal but the co-availability of the required signals: the method worked at 62% on our own devices and only 10% on third-party ones, and before that it had not worked at all, because both of its original pairs rested on a signal present in 2 of 149 records. We therefore present this

TABLE III

WHICH PROPERTIES EACH METHOD HAS. THE THREE RIGHTMOST COLUMNS ARE THE PROPERTIES THIS WORK SET OUT TO OBTAIN TOGETHER (SECTION I). CELLS ARE QUALITATIVE: THE METHODS REPORT RESULTS ON DISJOINT DEVICE SETS, SO NO ROW IS COMPARABLE TO ANOTHER AS A SCORE, AND NONE IS GIVEN HERE. “INTERACTION COST” MEANS RADIO ACTIVITY BEYOND PASSIVE DISCOVERY. THIS WORK IS THE LAST ROW.

Method	Signals it reads	Low interaction cost	Distinguishes many OSes	Reports conflicts between signals
Kavisankar <i>et al.</i> [1] (flooding the target)	Responses under induced stress; timing	No: floods the target and changes its state	Yes: OS and version	No: one verdict, always emitted
Celosia and Cunche [2] (single protocol field)	GATT attribute profile	Yes: one read, device undisturbed	No: device model, not an OS	No: a uniqueness score, not a disagreement
Santana-Cruz <i>et al.</i> [3] (radio layer only)	Radio-layer emissions	Yes in radio terms, but requires dedicated hardware	No: a device identity, not an OS	No: a confidence value only
Baseline scanner, this group (extended here)	CoD, SDP list, GATT UUIDs, OUI, LMP version	Medium: one connect-and-query pass	Yes: OS and version	No: all signals fused into one weighted score
This work	The baseline’s signals, plus the LMP feature bitmap, Modalias, GATT Appearance and connection parameters, and the full SDP tree	Yes: Layers 1–2 reuse the baseline’s connection or read the host cache; almost zero new connections	Yes: OS family and version, same branches as the baseline	Yes: explicit conflict flag and lowered confidence — but gated on signal co-availability (Section V)

TABLE IV

SIGNAL AVAILABILITY IN THE WILD. MODALIAS IS A HOST-CACHE READ RATHER THAN A LIVE SIGNAL, AND IS REPORTED AGAINST ALL ADDRESSES IN THE ROUND. COUNTS ARE GIVEN RAW: AT THESE SAMPLE SIZES A PERCENTAGE WOULD READ AS AN ESTIMATE IT CANNOT SUPPORT.

	Pilot ($n=18$)	Sweep ($n=52$)
Never reached by pipeline	16	46
Reached	2	6
Modalias present (of all addresses)	8/18	0/52

TABLE V

COVERAGE OF THE CLASS-OF-DEVICE VERSUS ADVERTISED-NAME PAIR, SPLIT BY WHOSE DEVICES WERE SCANNED. COVERAGE MEANS BOTH HALVES RESOLVED, SO THE PAIR COULD BE EVALUATED AT ALL.

Population	Coverage
Own devices	62%
Third-party devices	10%
All scans	48%

work not as an accuracy success but as a measurement that openly reports a constraint we observed.

The next steps follow directly from the non-goals stated in the introduction. An accuracy claim needs a ground-truth dataset, a layer-by-layer ablation, and a calibrated confidence function; each of those is future work, and none of them is assumed here.

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